Alberta Export Growth and Diversification

Prepared as Part of the Written Assessment Trade Data Scientist Recruitment: Question 1

Candidate Trade Data Scientist: Abeba N. Turi, Ph.D.
LinkedIn | Google Scholar
October 12, 2026

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Methodology

- Statistics Canada 2025 domestic exports were aggregated by Alberta origin, destination and HS4 product. The ten leading markets and five leading products in each market were ranked by Canadian-dollar export value (Attachments Q1-1 and Q1-2).
- The model training dataset uses 2024 BACI bilateral HS4 trade among Canada, the United States, Mexico, China, South Korea, Japan and the United Kingdom, with WDI and gravity indicators assembled as supporting factors. Alberta is treated separately; Canadian proxies are required where provincial covariates are unavailable.
- Five algorithms were compared using exporter-held-out validation, with Mexico and the United Kingdom held out. Results are judged using R^2, weighted absolute percentage error and log-scale RMSE (Attachment Q1-3).
- The ten-market 2025 allocation is a conditional global-demand scenario, not a validated machine-learning prediction. It is provided for directional comparison only, separately from observed 2025 exports (Attachment Q1-4).

Analysis

- Alberta exported CAD 152.04 billion in domestic merchandise to the United States in 2025, far ahead of China at CAD 10.19 billion. Japan followed at CAD 2.39 billion. The full ten-market ranking is in Attachment Q1-1.
- Crude petroleum (HS 2709) and petroleum gases (HS 2711) account for a large portion of U.S.-bound trade. The 50 destination-product combinations, including HS4 values, appear in Attachment Q1-2.
- The preferred histogram gradient boosting benchmark has weak out-of-exporter transferability: held-out original-scale R^2 is approximately 0.04 and WAPE approximately 0.95. These scores do not justify precise export forecasts (Attachment Q1-3).
- The separate global-demand scenario identifies the United States, China, India and Japan as leading conditional destinations. The scenario reflects assumed supply and demand shares, not observed shipments or model-validated export potential (Attachment Q1-4).
- Diversification planning should prioritize independent checks of demand, transport costs, trade barriers and Alberta production capacity. A reliable model-based top-ten 2025 forecast remains an outstanding assessment requirement; no unsupported forecast is claimed here.
- Economic implications: Heavy reliance on the U.S. and energy exports makes Alberta sensitive to market and commodity shocks. Targeted market development and infrastructure planning could improve resilience.
- Direction for future Work: Test the strongest opportunities against firm-level export readiness and validate 2025 predictions using additional trade periods.